Did They Grow Up With the Game? Directors' Age and Critics' Scores for Films Based on Video Games, 1993–2025

Course example · ECON 3916 Midterm Project · Dr. Richeng Piao

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Question	Among English-language films based on video games, 1993–2025, are films whose directors were younger when the game came out rated better by critics (the Rotten Tomatoes score), holding the release year and the format constant?
Hypothesis	A director who was young when the game came out is more likely to have grown up playing it, so they know what its players care about and should make a better film of it. I expect a negative coefficient on that age.
Headline finding	A director who was ten years younger when the game came out goes with a score 3.0 points higher (95% CI −1.9 to +7.9 points, N = 57). Without controls the association is clear. Once the release year is held fixed the estimate shrinks and its interval includes zero, and a median regression gives the same sign, also with an interval through zero. The hypothesis is neither supported nor rejected.
Data	One row is one film (57 in the sample, from 68 listed). Scores: Wikipedia's "List of films based on video games", citing Rotten Tomatoes, downloaded 2026-11-03. Game release year: the first release year on the Wikipedia page of the game or series each film is based on. Directors' birth dates: Wikidata.
Method	OLS of the Rotten Tomatoes score (percent of critics' reviews that are positive) on the director's age when the game came out, the release year and an animated indicator, with robust (HC1) standard errors.

1. The Question and the Data

Video-game adaptations have a bad record with critics, and for years the usual explanation was that the people making them had never played the games. That excuse is getting harder to make: many directors hired in the last decade were children when the games they adapt first came out. If a director who grew up with a game makes a better film of it, a studio choosing a director has a reason to look for one. The question matters to the studios making that choice and to the players who wait for these films.

My question is whether, among English-language films based on video games released from 1993 to 2025, films whose directors were younger when the game came out are rated better by critics, holding the release year and the format constant. The outcome is the film's Rotten Tomatoes score: the percent of critics' reviews that are positive. The main explanatory variable is the director's age when the game came out: the first release year of the game or series the film is based on, minus the director's birth year. My hypothesis is that a director who was young when the game came out is more likely to have played it, so I expect a negative coefficient. I chose this over the director's age at release because two directors of the same age can be adapting a game from their childhood or one from last year.

The data are a product of several measurement decisions. The list of films comes from Wikipedia's "List of films based on video games", which cites Rotten Tomatoes for every score. Rotten Tomatoes decides

which critics count, so a score of 30% means three in ten counted reviews were positive, not a mark out of 100. The score comes from critics rather than from the players the hypothesis is about, and when only a handful of critics reviewed a film it jumps around a lot. Checking the 5 most recent films against Rotten Tomatoes the next day, every score matched and no review count was off by more than 1. When each game came out comes from Wikipedia too: each film's infobox names the game or series it is "based on", and that page gives the first release year. Directors' birth dates come from Wikidata. Budgets, used only in Section 3, are put in 2024 prices with the BLS consumer price index (CPIAUCSL) from FRED.

Getting from the web page to a usable table took four steps, and each one changed the data. The page lists 68 films, 55 live-action and 13 animated. The title is not a key: Mortal Kombat, Resident Evil and Street Fighter were each remade, so I identify a film by its title and release year, and that key matters again when the game years are joined: the 1994 Street Fighter is based on Street Fighter II (1991), the 2026 one on the 1987 original. The scores arrive as text ("26% (46 reviews)[4]"), which I split into the score and the number of reviews. Merging the films with the birth dates on the director's name first matched only 58 of 68 films, because films with two directors came out of the web table with the names glued together ("Rocky MortonAnnabel Jankel"); matching on the first director listed fixed that. The game years matched 67 of the 68 films. The dating rule is written down: a franchise page's "origin" year, else its first release, else the earliest year in the game's release block. The rule matters twice. The 2018 Tomb Raider is based on the 2013 reboot, so it dates to 2013, not 1996. And Mario dates to 1981, because the franchise page notes that Mario first appeared in Donkey Kong; I keep that, since a director who played Donkey Kong knew Mario, but Super Mario Bros. (1985) would be a defensible alternative.

Two kinds of missing data affect the answer. 4 films have no score (Far Cry, Dead Trigger, Heavenly Sword, Scarygirl). Rotten Tomatoes posts no score below a minimum number of reviews, so whether a score exists depends on the review count, which I observe: the gap is not MCAR, and MAR given the review count is the defensible label. A film so weak that critics skipped it would also end up here, which would make the gap MNAR in the score itself. I cannot test that. Either way, my sample is the films critics bothered to review. I checked all four on Rotten Tomatoes itself: none shows a critics' score, and the percentages some pages quote are audience scores. Second, Wikidata had no birth date for 8 directors. Before labelling that gap I searched again and found 5: a redirect my first lookup did not follow, a director who shares a Wikipedia page with a co-writer, a film's second-listed director, and two names the table did not link. One name search found the wrong person, a politician with the same name, which is why every find in data/second_search.csv records its source. After the search only 1 rated film lacks a director's age, Moshi Monsters: The Movie; one film cannot show a mechanism, so I name it and leave it out. The same search converted the one budget given in pounds at the 2013 exchange rate instead of dropping it. Films released in 2026 are dropped by a rule written before fitting any model, since their reviews are still coming in. Table 1 shows each step; Table 2 summarizes the 57 films that remain.

Table 1. From 68 listed films to the sample

Step	Films
Films listed	68

Released in 2026 (reviews still coming in)	−6
No Rotten Tomatoes score	−4
No birth date for the director	−1
No release year for the source game	0
Films in the sample	57

Steps are applied in order, so a film missing two things is counted once, at the first step it fails.

Table 2. Summary statistics, 57 films, 1993–2025

Variable (units)	Mean	Median	SD	Min	Max	N
Rotten Tomatoes score (% positive reviews)	33.6	29.0	24.3	0.0	100.0	57
Reviews counted	128.8	120.0	85.2	5.0	327.0	57
Director's age when the game came out (years)	31.1	32.0	10.5	1.0	53.0	57
Director's age at release (years)	44.9	44.0	7.7	30.0	68.0	57
Release year	2011.8	2012	9.1	1993	2025	57
Budget ($m, 2024)	85.5	74.1	59.1	7.5	251.7	53

Budget in millions of 2024 dollars (CPI-U); 53 of the 57 films report one in dollars. Sources: Wikipedia (citing Rotten Tomatoes, Box Office Mojo and The Numbers), Wikidata, FRED. Downloaded 2026-11-03.

2. Descriptive Evidence

Most of these films are reviewed badly. The median score is 29.0%, the mean 33.6%, and only 9 of the 57 films pass 60% (Figure 1). The distribution is skewed right (skewness 0.72), which is why Table 2 reports the median beside the mean. A score is already a percentage between 0 and 100, so I model it directly rather than in logs, and check the result with a median regression in Section 4.

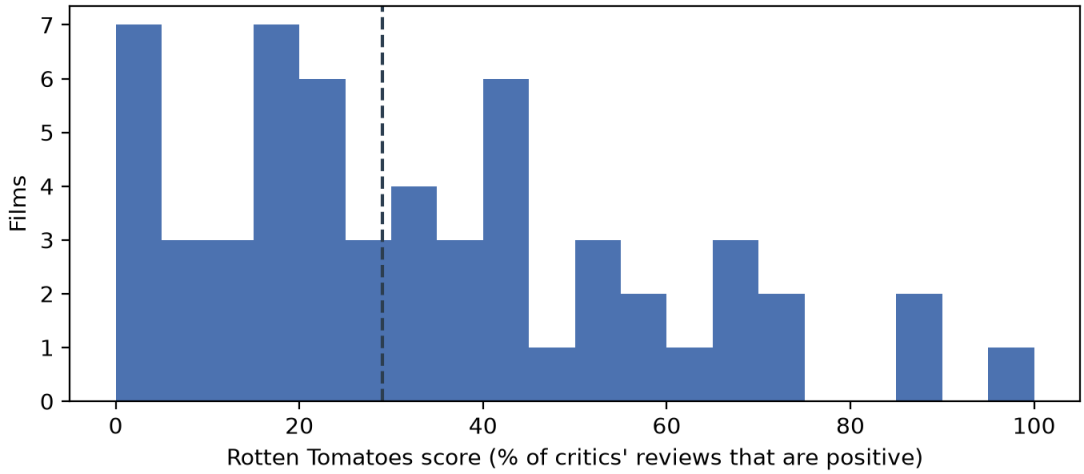


Figure 1. Most video-game adaptations are reviewed badly: the median film (dashed line) has 29.0% positive reviews, and only 9 of the 57 pass 60%. Source: Rotten Tomatoes scores as cited by Wikipedia.

Figure 2 plots each film's score against the director's age when the game came out, which runs from 1 to 53 years. The cloud tilts down: films by directors who were younger when the game appeared are rated better. Figure 3 is the reason I do not trust that raw pattern on its own. Scores rise with the release year: films released from 2016 on average 48.1%, against 19.9% for 1993–2005 and 24.1% for 2006–2015. And the recent films adapt games that came out when their directors were younger: the director's age when the game came out averages 34.3 years for films released before 2016 and 26.9 for films from 2016 (correlation with the release year -0.29). Part of the tilt in Figure 2 is the release year.

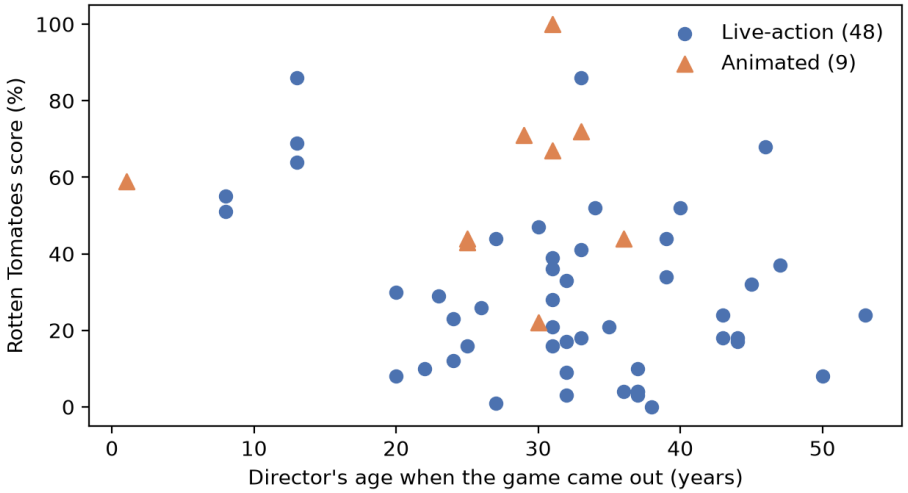


Figure 2. Films by directors who were younger when the game came out are rated better, before any controls; the youngest directors were about 1 when their game appeared. Source: as Figure 1; game years from Wikipedia, birth dates from Wikidata.

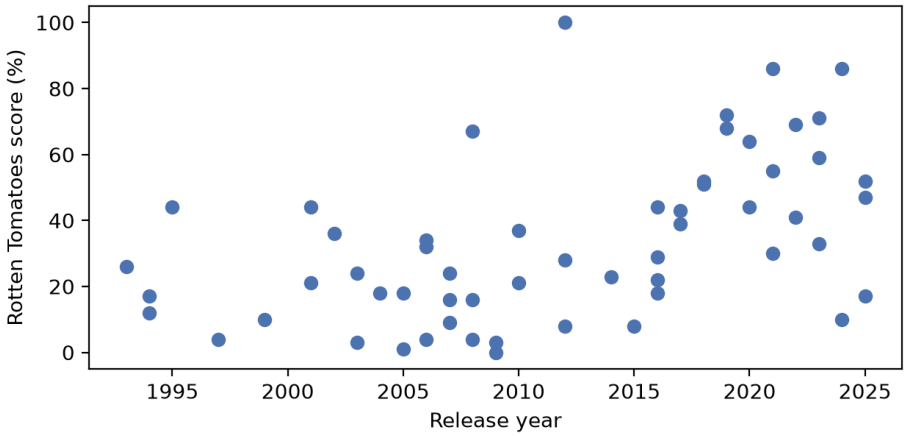


Figure 3. Recent adaptations are rated far better: films from 2016 on average 48.1%, against 19.9% to 24.1% before. Source: as Figure 1.

3. The Model

I fit the model below by OLS, with heteroskedasticity-robust (HC1) standard errors:

$$Score_i = \beta_0 + \beta_1\, AgeWhenGameCameOut_i + \beta_2\, Year_i + \beta_3\, Animated_i + u_i$$

The full results are in Table A1 in the appendix; column 2 is the main model. Without any controls (column 1) the coefficient on age when the game came out is −0.72 points per year (95% CI −1.17 to −0.27), an interval that excludes zero. Holding the release year and the format fixed (column 2), it is −0.30 points per year (standard error 0.25): a director who was ten years younger when the game came out goes with a score 3.0 points higher, with a 95% confidence interval from −1.9 to +7.9 points. The point estimate still has the sign my hypothesis predicts, but the interval now includes zero and even a small effect in the other direction. With 57 films I cannot tell these apart.

The second coefficient I interpret is the release year: +1.08 points per year (95% CI +0.50 to +1.67). Holding the director's age when the game came out and the format fixed, a film released ten years later scores about 11 points higher, and this interval excludes zero. The animated indicator is +23.4 points (95% CI +6.4 to +40.3), though only 9 films are animated. The R^2 is 0.40, against 0.25 without the release year.

Figure 4 draws the assumptions behind the model, and each control has a role in it:

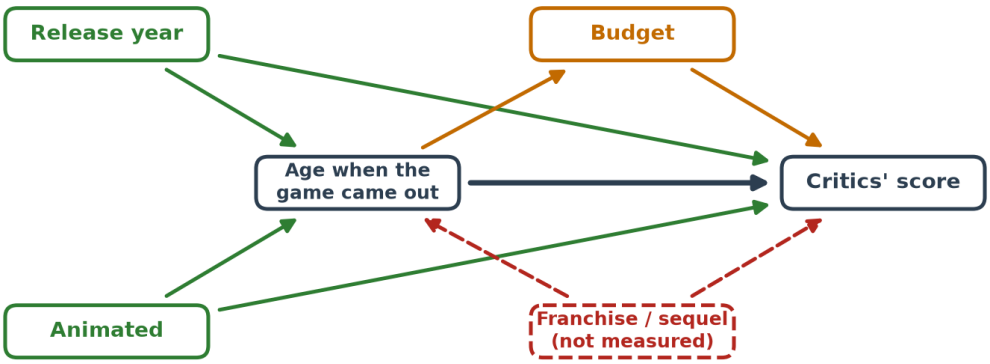


Figure 4. The causal graph I assume. Release year and format point into both the director's age when the game came out and the score, so the model controls for them. Budget could sit between the director and the score, so it stays out; Section 3 shows it is not a mediator in these data. Franchise and sequel status, which I cannot measure, leaves a path open (Section 5).

- **Release year** is a confounder, and in this sample it changes the answer. Scores rose sharply over three decades (Figure 3), and recent films adapt games that came out when their directors were younger: X averages 34.3 years for films before 2016 and 26.9 from 2016 (correlation -0.29), while each later year adds 1.08 points. Leaving it out would credit the director with the improvement of the whole genre. That is why the coefficient shrinks from −0.72 to −0.30 when the year goes in. It would be a bad control only if the director's age when the game came out could change the film's release year, which it cannot.
- **Animated** also blocks a confounding path: animated adaptations are made by different directors and reviewed on a different curve. The studio picks the format before hiring the director, so the director's age cannot cause it.
- **Budget** is left out on purpose. A studio may give a bigger budget to a director it trusts to know the game, and the money can change how good a film is, so budget would be a mediator and a bad control for my question. In these films it barely matters: the director's age when the game came out is almost unrelated to the budget (+0.2% per year, holding year and format fixed), and columns 3

and 4, on the same 53 films, give −0.36 and −0.37. In these data budget is not a mediator. The number of reviews is not a control either, because it depends on how much attention the film drew.

4. Diagnostics

I checked the main model with the tests from Topic 14. The residual plots in Figure A1 show no curve against the fitted values or against age when the game came out, so a straight line is a reasonable shape. The residual standard deviation is 19.4 points: critics' scores vary a great deal around the model. To test the shape directly I added age squared. Its coefficient is indistinguishable from zero ($p = 0.28$), the AIC rises from 503.6 to 504.3, and the prediction error on held-out films (10-fold cross-validation, repeated 20 times) rises from 20.24 to 21.15 points. The extra term buys nothing in the sample and does slightly worse out of it, which is what fitting noise looks like, so I keep the straight line.

Constant variance: the Breusch–Pagan test gives $p = 0.33$, no evidence of unequal variance. With 57 films the test has little power, so I lean on the plots and report robust standard errors anyway. The choice is visible in the age coefficient: its standard error is 0.26 with classical errors (95% CI −0.82 to +0.22) and 0.25 with HC1 (−0.79 to +0.19). Both include zero, so the conclusion does not depend on it.

Collinearity: the variance inflation factors are 1.12 for age when the game came out, 1.11 for the release year and 1.04 for the animated indicator. Age and year are related (Section 2), but not enough to matter; the wide interval comes from the small sample and the noisy scores.

Influence: using Cook's distance with the cutoff $4/n = 0.070$, 6 films stand out. The three largest are Resident Evil: Damnation (0.211), Ratchet & Clank (0.125) and Sonic the Hedgehog 3 (0.110). Resident Evil: Damnation's score of 100% rests on 5 reviews. A share from five critics is mostly noise. Without the 6 films reviewed fewer than 20 times (Double Dragon, Tekken, Resident Evil: Degeneration, Resident Evil: Damnation, Resident Evil: Vendetta, Resident Evil: Death Island), the coefficient is −0.34 points per year (standard error 0.25, $N = 51$).

Three more checks, all planned before fitting. A median regression, because the score is skewed, gives −0.22 points per year (95% CI −0.98 to +0.54): the same sign, with an interval that includes zero. Using the director's age at release instead, the cruder X, gives −0.25 (95% CI −0.86 to +0.37). And column 5 splits X into its two parts. Holding the release year fixed, X is the director's age at release minus the years the game had been out (13.8 on average). Estimated separately, age at release gets −0.28 (95% CI −0.89 to +0.33) and the game's age gets +0.33 (95% CI −0.50 to +1.15). Both point the way the X result does and neither interval excludes zero, so the data cannot tell a young director from an old, much-loved game.

5. Threats and Conclusion

For the coefficient to be causal, nothing else that moves a film's score could vary with the director's age when the game came out once the year and the format are fixed. Franchises are the clearest problem. The films whose directors grew up with the game are recent entries in big, carefully managed series (Sonic, Mortal Kombat, Mario), and studios hand those to directors they already trust. My data have no franchise or sequel variable, so that path stays open. The second problem is measurement, and part of it is built into X itself. A low X can be a director who grew up with the game, or an old game adapted many

years later, and long-running series (Mario, Street Fighter, Mortal Kombat) are also the ones with the biggest fan bases; column 5 cannot separate the two. Age when the game came out is also only a stand-in for having played it: a director of 12 may never have owned a console, and one of 35 may have played it every night.

One result I found along the way shows how easy it is to fool yourself. Directors who were under 20 when the game came out score 24.5 points higher, with an interval that excludes zero. I had not planned that cut, and it rests on 6 films (Rampage, Mortal Kombat, three Sonic films and The Super Mario Bros. Movie), all recent adaptations of old franchise games. What it picks up is franchise and timing rather than familiarity with the game, and I report it only to show what a subgroup like that looks like.

So the estimate is a conditional association, not the effect of having grown up with the game. Among these 57 films, holding the release year and the format fixed, a director who was ten years younger when the game came out goes with a score 3.0 points higher, with an interval from −1.9 to +7.9 points. The hypothesis is neither supported nor rejected: the raw association is clear, most of it goes with the release year, and what is left has the predicted sign in OLS and the same sign in a median regression, with intervals that include zero. With another month I would code each director's own gaming history from interviews, add a franchise flag, weight each score by its review count, and add audience scores to see whether players and critics agree.

References

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Appendix

Table A1. Rotten Tomatoes score regressed on the director's age when the game came out and controls

	(1) Age only	(2) Main	(3) Main, budget films	(4) Adds budget	(5) X split
Age when the game came out (years)	-0.72	-0.30	-0.36	-0.37	
	(0.23)	(0.25)	(0.24)	(0.23)	
Age at release (years)					-0.28
					(0.31)
Years the game had been out					0.33
					(0.42)
Release year		1.08	1.12	1.11	1.06
		(0.30)	(0.31)	(0.32)	(0.38)
Animated (0/1)		23.36	12.47	11.10	23.29
		(8.67)	(8.44)	(7.71)	(8.73)
log budget				4.53	
				(4.96)	
Constant	55.91	-2138.94	-2218.45	-2207.50	-2104.08
	(7.80)	(603.99)	(629.15)	(645.64)	(765.77)
N	57	57	53	53	57
R^2	0.10	0.40	0.36	0.38	0.40

Note: Dependent variable: Rotten Tomatoes score, percent of critics' reviews that are positive. OLS with robust (HC1) standard errors in parentheses. Column 2 is the main model. Column 3 re-estimates it on the 53 films with a dollar budget, so that column 4, which adds the log budget (Section 3), compares the same films. Column 5 splits X into its two parts, the director's age at release and the years the game had been out when the film was made (X = the first minus the second).

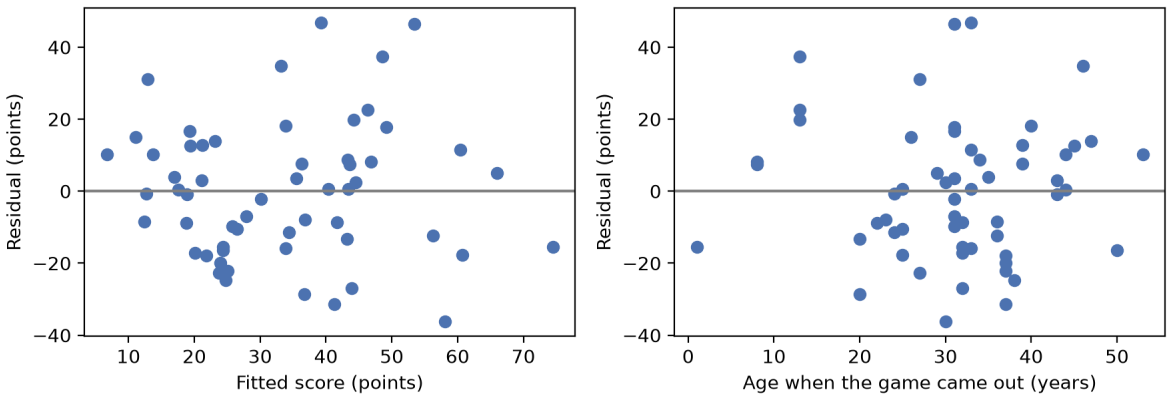


Figure A1. The main model's residuals show no curve against the fitted values (left) or the director's age when the game came out (right); the spread is wide throughout.

Reproducing the numbers. Every number in this report is printed by the last code cell of the notebook in the repository, "Numbers my report quotes". The README gives the download step and the order to run the cells.